Applied Business Analytics

Group Assignment

Total: 40 Marks

Based on the case study "Predicting Earnings Manipulation by Indian Firms using Machine Learning Algorithms", answer the following questions:

1. The number of manipulators is usually much less than non-manipulators. What kind of modeling problems one can expect when cases in one class are much lower than the other class in a binary classification problem? How can we handle such problems?

   2 Marks

2. Using the data provided, develop a logistic regression model that can be used by MCA Technologies Private Limited for predicting probability of earnings manipulation.

   14 Marks

3. Comment on the model developed. Discuss accuracy/ goodness of the model and the insights derived.

   10 Marks

4. Develop other ML model(s) to predict probability of earning manipulation. Interpret your findings.

   8 Marks

5. Based on the findings from the models developed, recommend a deployment strategy that should be adopted by MCA Technologies.

   6 Marks

Deliverables:

A. One professionally written case study report per group, inclusive of the following:

1. A cover page with project title, team members' names, roll number.
2. One-page summary highlighting main features of your analysis, modelling, and insights based on results obtained.
3. Main case study report, not exceeding 20 pages, including a concise understanding of the problem, regression model(s), graphical representations (residual plots, P-P plots, etc.), results, and conclusions. Specific recommendations for the different questions listed above must also be included in the report.
4. Assignment should be prepared on A4/Letter single-sized pages, 12 point Times New Roman font, and 1.5 line spacing.

B. Python Script used for the analysis